

The Fractional Fourier Transform on Graphs

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Abstract—The emerging field of signal processing on graphs merges algebraic or spectral graph theory with discrete signal processing techniques to process signals on graphs. In this paper, a definition of the fractional Fourier transform on graphs (GFRFT) is proposed and consolidated, which extends the discrete fractional Fourier transform (DFRFT) in the same sense the graph Fourier transform (GFT) extends the discrete Fourier transform (DFT). The definition is based on the eigenvalue decomposition method of defining DFRFT, for it satisfies all the agreeable properties expected of the discrete fractional Fourier transform. Properties of the GFRFT are discussed, and examples of GFRFT of some graph signals are given to illustrate the transform.

I. INTRODUCTION

Data with high-dimensional structures residing on the vertices of weighted graphs are being generated at an increasing rate in a wide range of applications, such as social, economic, transportation, energy, sensor, biological networks and so on. This large set of data, compared to ordinary time-series or images, has complex structures and calls for new types of processing techniques, which ultimately leads to the emerging field of *signal processing on graphs* [1], [2].

Signal processing on graphs extends the conventional *discrete signal processing* (DSP), by modelling the underlying structure of the signal by a graph, and indexing the signal by vertices. Related work covers graph-based transforms [3], [4], [5], [6], filtering [3], [7], [8], sampling [9], [10], [11], reconstruction and recovery [12], [13], and so on.

There are two fundamental frameworks of signal processing on graphs, based on spectral and algebraic approaches respectively. Both of the frameworks define fundamental notions and concepts for the theory, yet their disparate bases contribute to different definition and techniques for analysing graph signals.

The first one derives from *spectral graph theory*, which uses graph Laplacian matrix as its basic building block [1]. It expands the graph signal into the eigenfunctions of the Laplace operator to define the graph Fourier transform, and the corresponding spectrum is represented by the eigenvalues. This approach, however, is limited to undirected graphs only, as the standard graph Laplacian matrix is required to be symmetric and positive semi-definite.

The second one is called *discrete signal processing on graphs* (DSP_G) [2], [3], which originates from the *algebraic signal processing* (ASP) and uses adjacency matrix (the graph shift operator) as its basic building block. Particularly, in this framework, the graph Fourier transform expands the signal onto eigenvectors of the adjacency matrix, and defines the spectrum by the corresponding eigenvalues. As the graph shift operator, or the adjacency matrix, is not restricted to be symmetric, the second approach can be applied to arbitrary graphs.

An important technique to investigate time-series or graph signals is to adopt appropriate transforms. For conventional signal processing, Fourier analysis is one of the most basic and the most commonly used. Beside the Fourier transform per se, joint time-frequency analysis tools, including the Wigner distribution [15], the short-time Fourier transform (STFT) [15], [16] and the continuous wavelet transform (CWT) [17] are frequently used in speech processing, radar, or quantum physics.

The fractional Fourier transform (FRFT) was proposed in 1980 [18], and later introduced to the signal processing community in 1994 [19]. It is a set of linear transformations that generalizes the Fourier transform, and can convert a signal into any intermediate domain between time and frequency. The FRFT can also be interpreted as expanding a signal onto a basis of linear frequency modulation signals. Its applications cover a wide range, from filter design to pattern recognition.

Before introducing the fractional point of view into the field of DSP_G, it needs to be discretized for DSP. Since the discretization of the fractional Fourier transform cannot be obtained by directly sampling in time domain, the discretization of the FRFT has been widely studied. There are three mainstream types of method to defining and calculating the discrete fractional Fourier transform (DFRFT): linear combination-type [20], sampling-type [21] and eigendecomposition-type [22], [23], [24]. The different methods have been studied and compared in details in [25].

In this paper, a definition of the fractional Fourier transform on graphs (GFRFT) is proposed and consolidated by the following steps. In Section II, the theories on which the study is based are briefly reviewed. Section III presents the definition of GFRFT and several properties. In Section IV, examples of graph signals under GFRFT are given to illustrate the

This work is sponsored by the National Natural Science Foundation of China (No. 61671063).

transform. Finally, Section V concludes the paper.

II. PRELIMINARIES

A. Graph Fourier transform

In this study, the DSP_G (or ASP) framework is adopted, and the following reviews some of the most fundamental notions and concepts. A detailed introduction to the theory can be found in [2], [3].

For a dataset where data elements are connected to each other according to a certain relation, the relation is denoted by a graph $\mathcal{G} = (\mathcal{V}, \mathbf{A})$, where $\mathcal{V} = v_0, \dots, v_{N-1}$ is the set of *vertices*, and $\mathbf{A}$ is the weighted *adjacency matrix* of the graph. Each data element is indexed by a node v_n , and each weighted edge $\mathbf{A}_{n,m} \in \mathbb{C}$ from v_m to v_n denotes the relation of the m th data element to the n th one. If the graph is undirected, that is the relation is bidirectional, we have $\mathbf{A}_{n,m} = \mathbf{A}_{m,n}$, which means $\mathbf{A}$ is symmetrical.

Based on this graph, the dataset is referred to as graph signal, which is defined as a map

$$\mathbf{s} : \mathcal{V} \rightarrow \mathbb{C}, \quad (1)$$

$$v_n \mapsto s_n, \quad (2)$$

and could be written as a complex-valued vector

$$\mathbf{s} = [s_0 \ s_1 \ \dots \ s_{N-1}]^T \in \mathbb{C}^N.$$

Parallel to the classical Fourier transform which expands the signal onto a basis of exponent signals, the graph Fourier basis is given by the Jordan eigenvectors of the adjacency matrix $\mathbf{A}$, and the distinct eigenvalues $\lambda_0, \lambda_1, \dots, \lambda_{M-1}$ are the *graph frequencies* that form the *spectrum* of the graph. Jordan eigenvectors corresponding to a frequency λ_m are called *frequency components*.

As Jordan eigenvectors form the matrix $\mathbf{V}$ that reduces $\mathbf{A}$ to its Jordan normal form

$$\mathbf{A} = \mathbf{V} \mathbf{J}_\mathbf{A} \mathbf{V}^{-1},$$

the *graph Fourier transform* of a graph signal can be defined as

$$\hat{\mathbf{s}} = \mathbf{F} \mathbf{s} := \mathbf{V}^{-1} \mathbf{s}, \quad (3)$$

where $\mathbf{F} = \mathbf{V}^{-1}$ is the graph Fourier transform matrix. The n th component $\hat{s}_n$ represents the *frequency component* of the signal $\mathbf{s}$.

The *inverse graph Fourier transform* is defined as

$$\mathbf{s} = \mathbf{F}^{-1} \hat{\mathbf{s}} := \mathbf{V} \hat{\mathbf{s}}, \quad (4)$$

The DSP_G framework is the extension of the classical DSP theory. Periodic time-series can be presented by a cycle graph shown in Fig. 1 [14].

The directed edges represent the flow of time from past to future. The adjacency matrix of a cycle graph is

$$\mathbf{A} = \mathbf{C} = \begin{bmatrix} & & & 1 \\ 1 & & & \\ & \ddots & & \\ & & 1 & \end{bmatrix}. \quad (5)$$



Fig. 1. Time-series with the length of N represented by a cycle graph.

This matrix is diagonalizable and the eigendecomposition is

$$\mathbf{C} = \frac{1}{N} \mathbf{DFT}_N^{-1} \begin{bmatrix} e^{-j \frac{2\pi}{N} 0} & & \\ & \ddots & \\ & & e^{-j \frac{2\pi}{N} 0} \end{bmatrix} \mathbf{DFT}_N \quad (6)$$

where $\mathbf{DFT}_N$ denotes the discrete Fourier transform matrix. In this case, as intuitively, the GFT matrix $\mathbf{F} = \mathbf{DFT}_N$.

It should be noted that the definition of GFT and its inverse depends on the choice of eigenvectors, which is unnecessarily unique. How to choose the eigenvectors, however, is beyond the scope of this paper. Rather, it is assumed they are fixed. Nevertheless, the desirable choice of the eigenvectors to optimize the analysis remains an open issue.

B. Continuous fractional Fourier transform

The continuous fractional Fourier transform [18], [19] can be viewed as a linear operator, which rotates the time axis to the frequency axis by an angle of ϕ (rather than $\pi/2$ as a Fourier operator) in the time-frequency plane, and the transformed signal would be a representation of the signal along the rotated axis u making an angle of ϕ with the time axis.

The a th-order FRFT is defined for $0 < |a| < 2$ through its integral kernel (Mehler kernel) as

$$\mathcal{F}^a f(u) := \int_{-\infty}^{\infty} K_a(u, t) f(t) dt \quad (7)$$

$$K_a(u, t) := K_\phi e^{j\pi(u^2 \cot \phi - 2ut \csc \phi + t^2 \cot \phi)}$$

where

$$\phi := a\pi/2,$$

$$K_\phi := \exp[-j(\pi \operatorname{sgn}(\phi)/4 - \phi/2)] / |\sin \phi|^{0.5}.$$

The kernel $K_a(u, t)$ is defined separately for $a = 0$ and $a = \pm 2$ as $K_0(u, t) := \delta(u - t)$ and $K_{\pm 2}(u, t) := \delta(u + t)$. The definition is extended outside the interval $[-2, 2]$ by noting that $\mathcal{F}^{4l+a} f(u) = \mathcal{F}^a f(u)$ for any integer l .

The Mehler kernel is known to have the spectral expansion

$$K_a(u, t) = \sum_{k=0}^{\infty} \psi_k(u) e^{-j \frac{\pi}{2} k a} \psi_k(t) \quad (8)$$

where $\psi_k(t)$ denotes the k th Hermite-Gaussian function, which is also the eigenfunction of Fourier transform.

The properties of FRFT include:

- 1) Unitarity: $(\mathcal{F}^a)^{-1} = \mathcal{F}^{-a} = (\mathcal{F}^a)^\dagger$, where $(\cdot)^\dagger$ denotes Hermitian conjugation;
- 2) Index additivity: $\mathcal{F}^a \circ \mathcal{F}^b = \mathcal{F}^b \circ \mathcal{F}^a = \mathcal{F}^{a+b}$;
- 3) Zero rotation: $\mathcal{F}^0 = \mathbf{Id}$;
- 4) Reduction to Fourier transform when $a = 1$: $\mathcal{F}^1 = \mathcal{F}$.

C. Discrete fractional Fourier transform

For a discrete fractional Fourier transform in the strict sense, it is supposed to have the following properties [23]:

- 1) Unitarity;
- 2) Index additivity;
- 3) Reduction to Fourier transform when $a = 1$;
- 4) Resemblance to the continuous FRFT.

It can be easily checked that all three mainstream types of DFRFT reduce to Fourier transform when $a = 1$, and their other properties are as listed in Table 1 [25]. Clearly, the eigendecomposition method satisfies all the desirable properties, so in the current study, a similar method is used to generalize the transform onto graphs.

Using eigendecomposition method, the DFRFT is defined as

$$\mathbf{F}^a[m, n] = \sum_{k=0}^{N-1} u_k[m] e^{j \frac{\pi}{2} k a} u_k[n], \quad (9)$$

denoting the discrete Hermite-Gaussian series as $u_k[n]$. The definition of the discrete Hermite-Gaussian series is given by the common eigenvectors of two commutative matrix – those of the discrete Fourier transform and the discrete Laplacian operator, so that the DFRFT would resemble its continuous version. The $e^{j \frac{\pi}{2} k a}$ in the middle can be interpreted as the eigenvalue of DFT to the a th order.

III. THE FRACTIONAL FOURIER TRANSFORM ON GRAPHS

A. Definition

The definition generalizes the discrete fractional Fourier transform in the same sense the graph Fourier transform generalizes the discrete Fourier transform.

We first assume that the GFRFT can be expressed in the matrix form as

$$\hat{\mathbf{s}} = \mathcal{F}^a \mathbf{s} := \mathbf{F}^a \mathbf{s}. \quad (10)$$

To begin with a simple circumstance, consider a signal $\mathbf{s}$ on graph $\mathcal{G}$ with a diagonalizable adjacency matrix $\mathbf{A} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$. Assume that the GFT matrix $\mathbf{F}$ can be orthogonally diagonalized, i.e.

$$\mathbf{F} = \mathbf{V}^{-1} = \chi \mathbf{\Lambda} \chi^T = \sum_{k=0}^{N-1} \chi_k \lambda_k \chi_k^T, \quad (11)$$

where λ_k and χ_k denote the k th eigenvalue of $\mathbf{F}$ and its corresponding eigenvector respectively. Parallel to definition

of the DFRFT matrix to the a th order, the GFRFT matrix can be given by

$$\mathbf{F}^a[m, n] = \sum_{k=0}^{N-1} \chi_k[m] \lambda_k^a \chi_k[n]. \quad (12)$$

Then we can generalize the definition into a broader domain using Jordan decomposition. As mentioned previously, when the adjacency matrix is not diagonalizable, the GFT matrix can be defined by the Jordan decomposition of $\mathbf{A} = \mathbf{V} \mathbf{J} \mathbf{V}^{-1}$. Further, the GFT matrix is decomposed into its Jordan eigenvectors

$$\mathbf{F} = \mathbf{V}^{-1} = \mathbf{P} \mathbf{J} \mathbf{P}^{-1}, \quad (13)$$

and the definition of the GFRFT matrix would be

$$\mathbf{F}^a := \mathbf{P} \mathbf{J}^a \mathbf{P}^{-1}. \quad (14)$$

Note that the matrix function of the Jordan normal form can be computed by applying the function to each Jordan block, so that the elements of the k th super-diagonal of the resulting block are $\frac{f^{(k)}(\lambda)}{k!}$. Specifically, for the a th-order power function, the k th super-diagonal is $\binom{a}{k} \lambda^{a-k}$, where the binomial coefficients are defined as $\binom{a}{k} = \prod_{i=1}^k \frac{a+1-i}{i}$. For example,

$$\begin{bmatrix} \lambda_1 & 1 & 0 & 0 & 0 \\ 0 & \lambda_1 & 1 & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 \\ 0 & 0 & 0 & \lambda_2 & 1 \\ 0 & 0 & 0 & 0 & \lambda_2 \end{bmatrix}^a = \begin{bmatrix} \lambda_1 & \binom{a}{1} \lambda_1^{a-1} & \binom{a}{2} \lambda_1^{a-2} & 0 & 0 \\ 0 & \lambda_1^a & \binom{a}{1} \lambda_1^{a-1} & 0 & 0 \\ 0 & 0 & \lambda_1^a & 0 & 0 \\ 0 & 0 & 0 & \lambda_2^a & \binom{a}{1} \lambda_2^{a-1} \\ 0 & 0 & 0 & 0 & \lambda_2^a \end{bmatrix}.$$

It can be verified that this definition of matrix power function holds the index additivity $\mathbf{J}^a \mathbf{J}^b = \mathbf{J}^{a+b}$ [26].

Once we have defined the GFRFT matrix, the GFRFT to the a th order would then be

$$\hat{\mathbf{s}} = \mathcal{F}^a \mathbf{s} := \mathbf{F}^a \mathbf{s}. \quad (15)$$

Its inverse $(\mathcal{F}^a)^{-1}$ would be $\mathcal{F}^{-a}$, for

$$\begin{aligned} \mathcal{F}^{-a} \circ \mathcal{F}^a &= \mathbf{F}^{-a} \mathbf{F}^a = \mathbf{P} \mathbf{J}^{-a} \mathbf{P}^{-1} \mathbf{P} \mathbf{J}^a \mathbf{P}^{-1} \\ &= \mathbf{P} \mathbf{J}^{-a} \mathbf{J}^a \mathbf{P}^{-1} = \mathbf{P} \mathbf{P}^{-1} = \mathbf{I}. \end{aligned} \quad (16)$$

B. Properties

- 1) Zero rotation

$$\mathbf{F}^0 = \mathbf{P} \mathbf{J}^0 \mathbf{P}^{-1} = \mathbf{P} \mathbf{I} \mathbf{P}^{-1} = \mathbf{P} \mathbf{P}^{-1} = \mathbf{I}, \quad (17)$$

so the 0th-order GFRFT of a signal is the signal itself.

- 2) Reduction to GFT when $a = 1$

$$\mathbf{F}^1 = \mathbf{P} \mathbf{J}^1 \mathbf{P}^{-1} = \mathbf{P} \mathbf{J} \mathbf{P}^{-1} = \mathbf{F}, \quad (18)$$

so the 1th-order GFRFT of a signal is the GFT signal.

TABLE I
COMPARING THREE MAIN TYPES OF DFRFTs

Properties	linear combination	sampling	eigendecomposition
unitary	✓	✓	✓
index additive	✓	×	✓
resemble FRFT	×	✓	✓

- 3) Index additivity As mentioned earlier, the matrix power function is index additive, so we have

$$\begin{aligned}\mathbf{F}^a \mathbf{F}^b &= \mathbf{P} \mathbf{J}^a \mathbf{P}^{-1} \mathbf{P} \mathbf{J}^b \mathbf{P}^{-1} = \mathbf{P} \mathbf{J}^{a+b} \mathbf{P}^{-1} \\ &= \mathbf{P} \mathbf{J}^{a+b} \mathbf{P}^{-1} = \mathbf{F}^{a+b}.\end{aligned}\quad (19)$$

- 4) Reduction to DFRFT on cycle graphs

It has been discussed that when $\mathbf{A} = \mathbf{C}$, the GFT matrix and the DFT matrix are identical, so $\mathbf{F}$ can be diagonalized by the Hermitian-Gaussian vector. Then the GFRFT matrix will be the same as the predefined DFRFT matrix.

- 5) Convolution theorem

The convolution in the fractional domain on the graph can be given by

$$\begin{aligned}(f * g)[n] &:= \sum_{m=0}^{N-1} \hat{f}[m] \hat{g}[m] F^{-a}[n, m], \\ \text{or } \mathbf{f} * \mathbf{g} &:= \mathbf{F}^{-a}(\hat{\mathbf{f}} \circ \hat{\mathbf{g}}).\end{aligned}\quad (20)$$

Here, the $\circ$ denotes the Hadamard product of matrices. Hence, in this way, the convolution theorem holds

$$\widehat{\mathbf{f} * \mathbf{g}} = \hat{\mathbf{f}} \circ \hat{\mathbf{g}}. \quad (21)$$

- 6) Translation property

The translation operator $\mathcal{T}_i$ can be given by convolution with a δ_i vector, whose components are all 0 except $\delta_i[i] = 1$.

$$\begin{aligned}\mathcal{T}_i \mathbf{f} &:= \sqrt{N} \mathbf{f} * \delta_i = \sqrt{N} \mathbf{F}^{-a}(\hat{\mathbf{f}} \circ \hat{\delta}_i) \\ &= \sqrt{N} \mathbf{F}^{-a}(\hat{\mathbf{f}} \circ \mathbf{F}_i^a),\end{aligned}\quad (22)$$

where $\mathbf{F}_i^a$ denotes the i th column of the matrix $\mathbf{F}^a$. Then its GFRFT is calculated to be

$$\begin{aligned}\mathcal{F}^a(\mathcal{T}_i \mathbf{f}) &= \sqrt{N} \mathcal{F}^a \mathbf{F}^{-a}(\hat{\mathbf{f}} \circ \mathbf{F}_i^a) \\ &= \sqrt{N} \hat{\mathbf{f}} \circ \mathbf{F}_i^a,\end{aligned}\quad (23)$$

which corresponds to the time-delay property.

IV. EXAMPLES

Examples of bipolar rectangular signal on the cycle graph [14] and the Minnesota road graph [29] are given in Fig. 2 and Fig. 3 respectively to illustrate the GFRFT. We can have a sense for the rotation of domain from vertex to frequency by observing the variation of GFRFT with different order a . Specifically, the first example shows that the GFRFT reduce to the conventional DFRFT on the cycle graph.

The calculation is based on the Graph Signal Processing toolbox (GSPBox) [30] on MATLAB, with minor modifications for the DSP_G framework. The eigenvalues are calculated by the function eig() and sorted in the descending order.

V. CONCLUSIONS

To sum up, a generalization of the graph Fourier transform is presented in the paper, which is designated *fractional Fourier transform on graphs*, or abbreviated as GFRFT. The transform, depending on the order a , can be interpreted as a rotation by an angle $\phi = a\pi/2$ in the vertex-frequency plane. In particular, it holds the following properties:

- 1) When $a = 0$, the GFRFT is the identity operator.
- 2) When $a = 1$, the GFRFT reduce to the GFT.
- 3) When the graph's $\mathbf{A} = \mathbf{C}$, the GFRFT reduce to the conventional FRFT.
- 4) Two successive GFRFTs to the a th and b th order respectively, are equivalent to a single GFRFT to the $a + b$ th order.

Further, the current study could possibly open new areas in the field of signal processing on graphs, and lead to further research. Hopefully, useful processing techniques and specific applications will be developed with the fractional insight.

ACKNOWLEDGMENT

This work is supported by the National Natural Science Foundation of China (No. 61671063).

The author also wishes to acknowledge valuable suggestions from Jie-dong Jiang and Shi-ji Lyu. They are both with School of Mathematical Sciences, Peking University.

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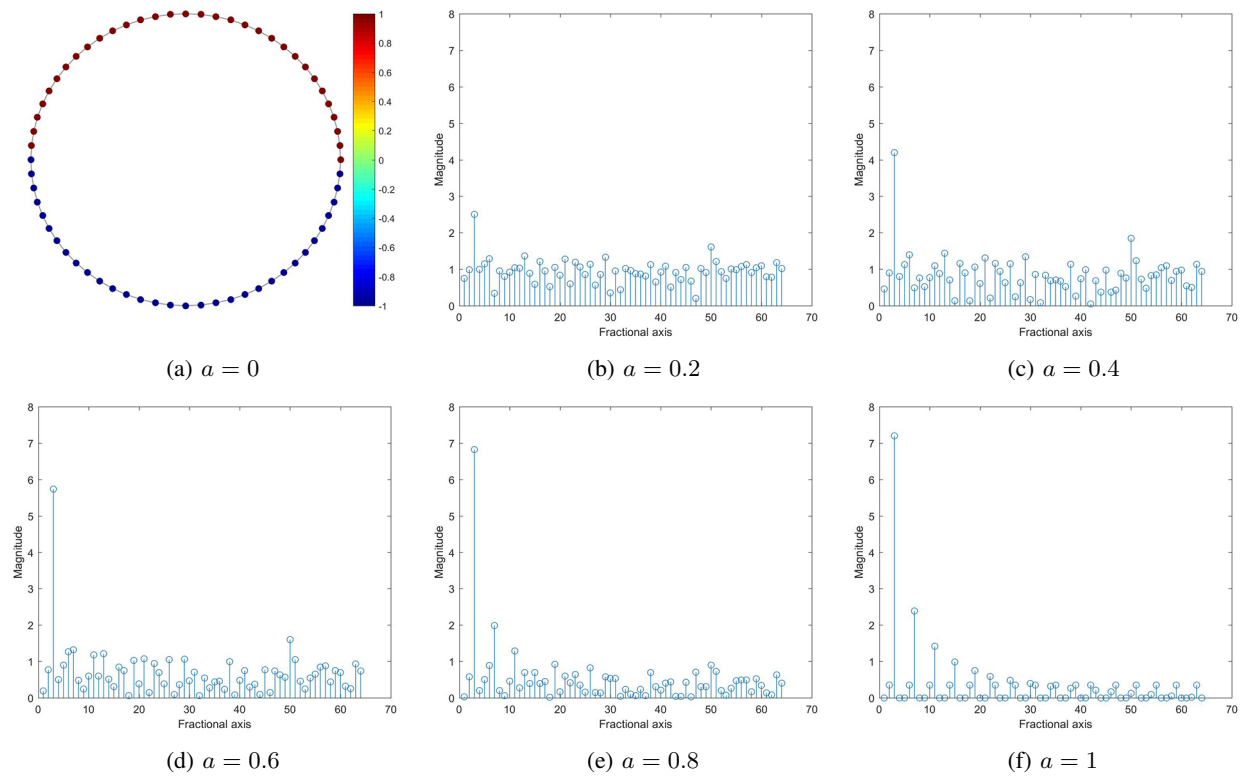


Fig. 2. The GFRFT of bipolar rectangular signal on the cycle graph with different a .

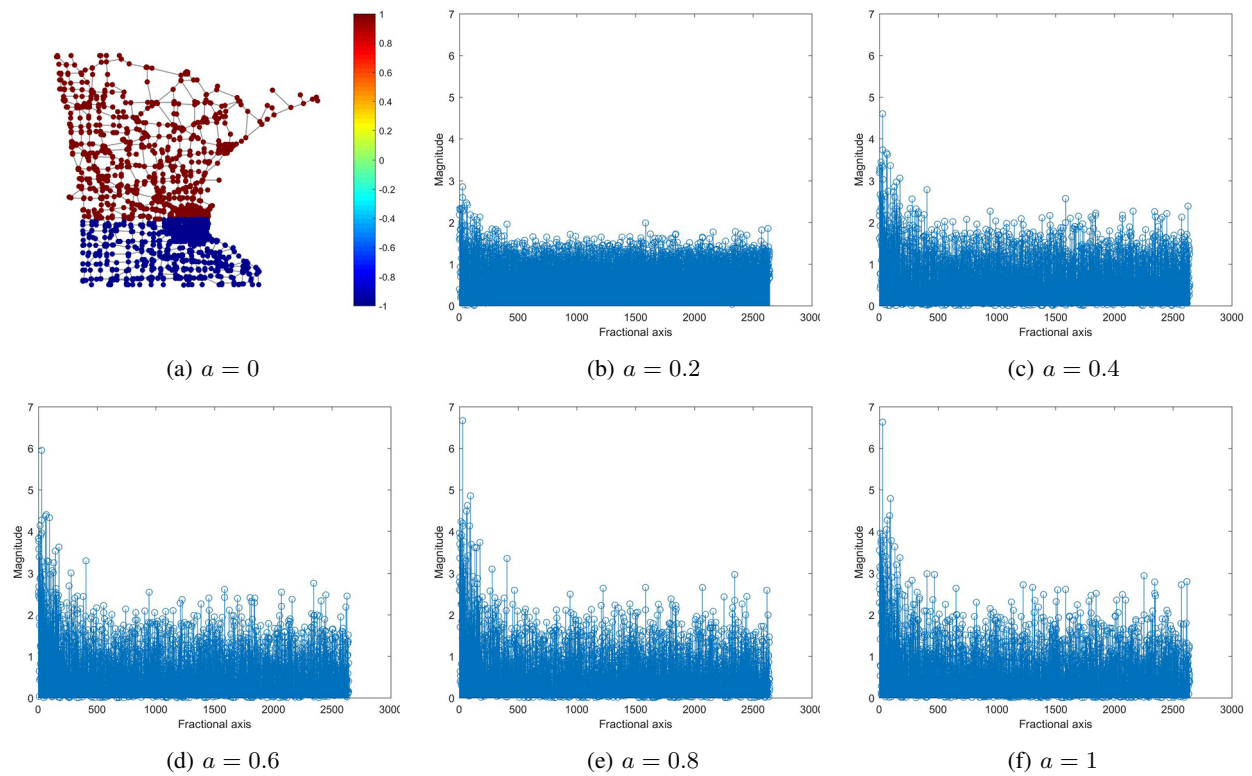


Fig. 3. The GFRFT of bipolar rectangular signal on the Minnesota road graph with different a .

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